



Distributed Systems

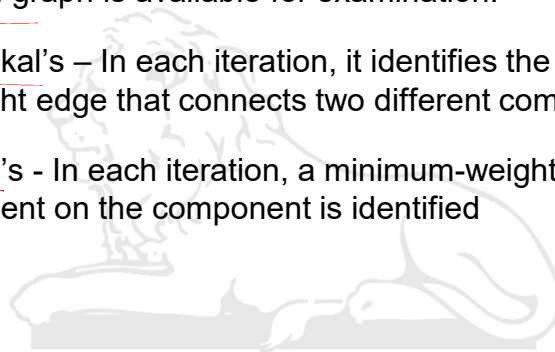
Elementary Graph Algorithms



MST



- Classical centralized MST algorithms assume that the entire weighted graph is available for examination.
 - Kruskal's – In each iteration, it identifies the minimum-weight edge that connects two different components,
 - Prim's - In each iteration, a minimum-weight edge incident on the component is identified



Minimum-weight spanning tree (MST) algorithm in a synchronous system



- In a distributed algorithm, each process can communicate only with its neighbors and is aware of only the incident links and their weights.
- It is also assumed that the processes know the value of $N = n$.
- The weight of each edge is unique in the network, which is necessary to guarantee a unique MST.
- A distributed algorithm by Gallagher, Humblet, and Spira (GHS) that generalizes the strategy of Kruskal's centralized algorithm

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Some Definitions



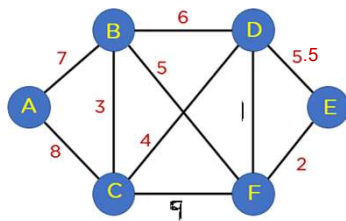
- A **spanning forest** of graph G is a subgraph G' of G
 - having the same node set as G and
 - a set of spanning trees, one spanning tree per “connected component” of G .
- All MST algorithms begin with a spanning forest having n nodes (or connected components) and without any edges.
- They then add a “minimum-weight outgoing edge” (MWOE) between two components.
- This forms a single spanning tree of the combined connected component together with MWOE.
- The addition of the MWOE is repeated until a spanning tree is produced for the entire graph (N, L)

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Kruskal's MST

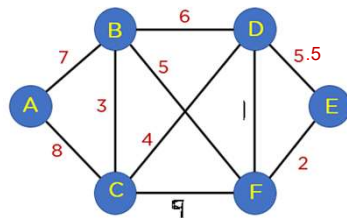


- Assume forest of graph components
- Maintain sorted list of edges
- In each of $n - 1$ iterations, identify minimum weight edge that connects two different components
- Include the edge in the MST

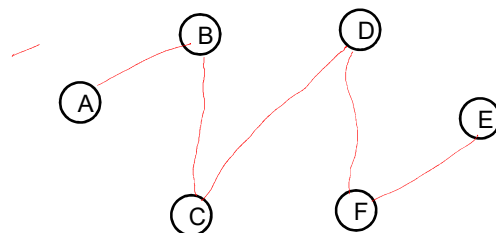


Source: <https://www.simplilearn.com/>

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DF	1	✓
FE	2	✓
BC	3	✓
CD	4	✓
BF	5	X
DE	5.5	X
BD	6	X
AB	7	



$n-1$

Spanning Tree



- Reason for Correctness of such algorithms:

Observation 5.1 For any spanning forest $\{(N_i, L_i) \mid i = 1 \dots k\}$ of a weighted undirected graph G , consider any component (N_j, L_j) . Let λ_j be the edge having the smallest weight among those that are incident on only one node in N_j . Then an MST for the graph G that includes all the edges in each L_i in the spanning forest, must also include edge λ_j .

- Intuitively, the logic is as follows:

For any component containing node set N_j , if edge x is used instead of the MWOE λ_j to connect with nodes in $N \setminus N_j$, then the resulting tree cannot be a MST because edge x can always be replaced with the MWOE that was not chosen to yield a lower cost tree.

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Synchronous Gallagher–Humblet–Spira MST Algorithm



- In a distributed algorithm, the addition of the edges should be done concurrently by having all the components identify their respective minimum-weight outgoing edge.¹
- Initially, each node is the leader of its component which contains only that node.
- The algorithm uses $\log(n)$ iterations.

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Synchronous Gallagher–Humblet–Spira MST Algorithm



SEARCH_MWOE (leader) // broadcast by current leader on
 // tree edges
 EXAMINE (leader) // sent on non-tree edges after
 // receiving SEARCH_MWOE
 REPLY_MWOE (local_ID, remote_ID) // details of potential
 // MWOEs are
 // convergecast to leader
 ADD_MWOE(local_ID, remote_ID) // sent by leader to add
 // MWOE and identify
 // new leader
 NEW_LEADER (leader) // broadcast by new leader after
 // merging components

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Synchronous Gallagher–Humblet–Spira MST Algorithm



leader = i;
 for round = 1 to $\log(n)$ do
 1. if leader = i then
 broadcast SEARCH_MWOE(leader) along marked edges of tree
 2. On receiving a SEARCH_MWOE(leader) message that was broadcast on marked edges:
 a) Each process i (including leader) sends an **EXAMINE** message along unmarked (i.e., non-tree) edges to determine if the other end of the edge is in the same component (i.e., whether its leader is the same).
 b) From among all incident edges at i, for which the other end belongs to a different component, process i picks its incident MWOE(localID, remoteID).
 3. The leaf nodes in the component initiate the **convergecast** using **REPLY_MWOEs**, informing their parent of their MWOE(localID, remoteID). All the nodes participate in this convergecast.

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Synchronous Gallagher–Humblet–Spira MST Algorithm



4. if leader = i then

await convergecast replies along marked edges.

Select the minimum MWOE(localID, remoteID) from all the replies.

broadcast ADD_MWOE(localID, remoteID) along marked edges of tree

// For process localID to mark the localID remoteID edge, i.e., include it in MST of component.

5. if an MWOE edge gets **marked by both the components** on which it is incident then

(a) **Define new_leader** as the process with the larger ID on which that MWOE is incident (i.e., process whose ID is $\max(\text{localID}, \text{remoteID})$).

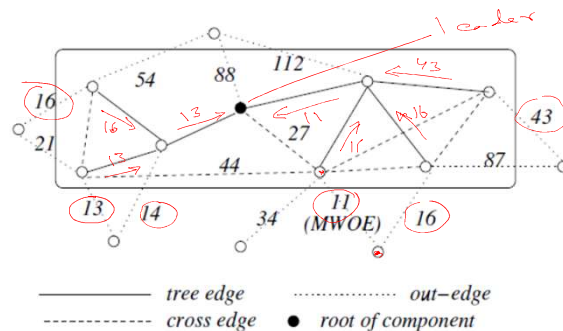
(b) *new_leader* identifies itself as the leader for the next round.

(c) *new_leader* **broadcasts NEW_LEADER** in the newly formed component along the marked edges announcing itself as the leader for next round.

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Gallagher–Humblet–Spira Algorithm



(a) root broadcasts SEARCH_MWOE; (b) convergecast REPLY_MWOE occurs; (c) root broadcasts ADD_MWOE; (d) if the MWOE is also chosen as the MWOE by the component at the other end of the MWOE, the incident process with the higher ID is the leader for the next iteration and broadcasts NEW_LEADER.

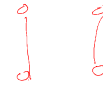
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Gallagher–Humblet–Spira Algorithm



- Each iteration goes through a broadcast–convergecast–broadcast sequence to identify the MWOE of the component, and to select the *leader* for the next iteration.



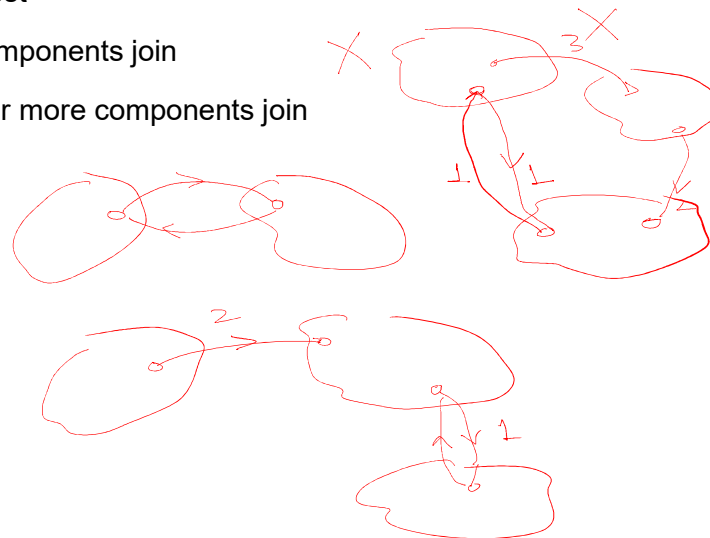
Correctness

- Each component picks exactly *one* MWOE with which it connects to another component.
- More than two components can join together in one iteration.
- If multiple components join, the resulting component is still a spanning forest
 - If two components join
 - If three or more components join

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- If multiple components join, the resulting component is still a spanning forest
 - If two components join
 - If three or more components join



Complexity



$n-1$

- Message Complexity:
 - In each of the $\log(n)$ iterations, $O(n)$ messages are sent along the marked tree edges.
 - There may be up to $l = |L|$ EXAMINE messages to determine the MWOEs in step 2 of each iteration.
 - Hence, the total message complexity is $O((n+l) \cdot \log(n))$.

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Leader election



- Algorithms such as the **minimum spanning tree** and **broadcast / convergecast** require a leader process that starts computation
- A leader is required in many distributed systems because algorithms are typically not completely symmetrical, and some process has to take the lead in initiating the algorithm
- Another reason is that we would **not want all the processes to replicate the algorithm initiation, to save on resources**.
- Leader election requires that all the processes agree on a common distinguished process (*leader*)

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Leader election



- A valid leader election algorithm must meet the following conditions:
- **Termination**: the algorithm should finish within a finite time once the leader is selected (for deterministic algorithms)
- **Uniqueness**: there is exactly one processor that considers itself as leader (even in presence of concurrent initiations)
- **Agreement**: all other processors know who the leader is.